PRACTICAL NO : 9

AIM: : A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

CODE:

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GNU nano 8.7

```c
#include <stdio.h>

int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frames[3];
    int total_pages = 12;
    int frames_count = 3;
    int i,j,k;

    int hit = 0, fault = 0;

    for(i=0;i<frames_count;i++)
        frames[i] = -1;

    for(i=0;i<total_pages;i++){
        int found = 0;

        for(j=0;j<frames_count;j++){
            if(frames[j] == pages[i]){
                hit++;
                found = 1;
                break;
            }
        }

        if(!found){
            int pos = -1;
            int farthest = i;

            for(j=0;j<frames_count;j++){
                int k;
                for(k=i+1;k<total_pages;k++){
                    if(frames[j] == pages[k])
                        break;
                }

                if(k == total_pages){
                    pos = j;
                    break;
                }

                if(k > farthest){
                    farthest = k;
                    pos = j;
                }
```

~

```
  GNU nano 8.7
        }

        if(!found){
            int pos = -1;
            int farthest = i;

            for(j=0;j<frames_count;j++){
                int k;
                for(k=i+1;k<total_pages;k++){
                    if(frames[j] == pages[k])
                        break;
                }

                if(k == total_pages){
                    pos = j;
                    break;
                }

                if(k > farthest){
                    farthest = k;
                    pos = j;
                }
            }

            if(pos == -1)
                pos = 0;

            frames[pos] = pages[i];
            fault++;
        }
    }

    printf("Total Page References = %d\n", total_pages);
    printf("Page Hits = %d\n", hit);
    printf("Page Faults = %d\n", fault);

    float hit_percent = (hit * 100.0) / total_pages;
    float fault_percent = (fault * 100.0) / total_pages;

    printf("Hit Percentage = %.2f%%\n", hit_percent);
    printf("Fault Percentage = %.2f%%\n", fault_percent);

    return 0;
}
```

OUTPUT:

~

```
Chanchal@DESKTOP-GARLJAI MSYS ~
# nano p9.c

Chanchal@DESKTOP-GARLJAI MSYS ~
# gcc p9.c -o p9

Chanchal@DESKTOP-GARLJAI MSYS ~
# ./p9
Total Page References = 12
Page Hits = 5
Page Faults = 7
Hit Percentage = 41.67%
Fault Percentage = 58.33%

Chanchal@DESKTOP-GARLJAI MSYS ~
#
```